

Question	Marking Guidance	Mark	Comments
08.1	Correct answer of 8.1 / 8.07/ 8.066 / 8.0658 / 8.06575 $\times 10^{-3}$ = 2 marks;; Incorrect answer but shows 8 / 8.1 / 8.07 / 8.066 / 8.0658 / 8.06575 = 1 mark Correct answer but not in standard form = 1 mark Incorrect rounding of correct answer in correct standard form e.g., 8.06 $\times 10^{-3}$ = 1 mark;	2 (2 x AO2)	Note. To award 2 marks $\times 10^{-3}$ is required. Accept 8×10^{-3} for 2 marks but 8.0×10^{-3} = 1 mark due to incorrect rounding.

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08.2	For (valid) comparison as rats vary in mass OR (So) each rat receives a quantity relative to their mass OR (So) <u>concentration</u> in the blood/body is the same;	1 (AO2)	Accept 'standardisation' for 'comparison'. Accept 'weight' for mass but ignore size.

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08.3	1. SDs do not overlap (for blood glucose concentration) OR SDs do not overlap (for mass); 2. So significant difference/increase (in blood glucose concentration) OR So significant difference/increase (in mass); 3. (Type II diabetes) causes high blood glucose (concentration); 4. Obesity/high body mass is a (risk) factor (in type II diabetes) OR High fat (diet) is a (risk) factor (in type II diabetes); 5. (Investigation) done on rats (not humans); 6. (Only shows) results after short-time period OR Long-term effects not known;	5 max (5 x AO3)	2. Accept 'difference/increase is not due to chance' for significant difference/increase. Ignore reference to stats test. 3. Accept 'associated/linked/have' for causes. 6. Accept 1, 2 or 3 weeks.

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08.4	1. (Type II) still produce/release insulin; 2. (Type II) cells/receptors less/not responsive/sensitive to insulin; 3. Pancreatic <u>cells</u> not destroyed (in type II diabetes); 4. Damage to pancreatic <u>cells</u> may affect processes/reactions (in the body);	2 max (2 x AO3)	1. Accept 'type 1 would not produce/release insulin' or this would cause type I diabetes. 2. Accept involves 'faulty receptors' or 'fewer receptors'. 3. Ignore pancreas is not destroyed.